

flea

NEWS 87 *August 2021*



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Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Editorial

More on Fleas as Keystone Parasites

Dear *Flea News* Reader,

Because of delays from the Covid-19 pandemic, this *Flea News* will contain citations from 2020. The 2021 citations should be in the next *Flea News*, thanks to everyone sending me material for *Flea News*.

In the last *Flea News*, the topic was "keystone fleas". In this editorial, I continue the discussion of keystone species effects, but I will emphasize fleas (and their pathogens) as keystone species, and describe a few examples.

Innumerable host associations of fleas with their hosts and pathogens are well-documented (Durden and Hinkle 2019). These associations may have great effects on community diversity, and on the resulting ecosystem function (Goldberg et al. 2020). The term "keystone fleas" was used to describe flea species *Megabothris abantis* and *Meringis hubbardi* that linked unique groups of mammals in the Great Basin Desert, but there was no attempt to determine how these "keystone fleas" might influence their community, for examples, by pathogen transmission or by selective parasitism that modulates strengths of competition, herbivory, or predation, and thereby changing community diversity (Bossard 2006).

Fleas affect many aspects of their host populations, including nesting behavior, survival rates, the number of young, and in birds, the size of egg clutches and hatching success. Hosts studied include cliff swallow (Brown et al. 1995), tree swallow (Rendell and Verbeek 1996), blue tit (with the flea *Ceratophyllus gallinae*, Richner and Tripet 1999, Bouslama et al. 2002, Fitze et al. 2004, Slomczyński et al. 2006), great tit (Richner et al. 1993, Oppliger et al. 1994, Heeb et al. 2000), starling (Mazgajski 1997), kit fox (Kluever 2019), gerbil (with the flea *Synosternus cleopatrae*, Hawlena et al. 2006), Cape ground squirrel (Hillegass et al. 2010), and Columbian ground squirrel (Neuhaus 2003).

Nonetheless, in regard to the Columbian ground squirrels (infested with *Oropsylla idahoensis* and *Oropsylla opisocroistis tuberculata* [sic]), Roth et al. (2019) declared: "Some subtle short term effects do manifest but are unlikely to culminate in detrimental long-term fitness consequences unless parasitemia is chronically sustained. As such, fleas, even when experimentally augmented in number to increase their impact, do not strongly affect these hosts. This result suggests that while hosts in poor

condition may exhibit high flea loads, flea infestation by itself is unlikely to debilitate hosts."

Similarly, Watson's (2013) meta-analysis of 38 datasets showed that wild vertebrate hosts' overall breeding success was not affected by their parasites, in spite of their significant impacts on other parameters of the host population. Surprisingly, flea infestations seem often to have negligible direct effects on their hosts, a finding reinforced by other studies (Warburton et al. 2020, Williams et al. 2020).

Sometimes, though, effects of flea parasitism are neither negative nor negligible, but indirectly positive for their hosts! This possibility seems often neglected because of the incorrect perception that parasites must harm their hosts and other species. These positive effects occur because fleas and their microbial symbiotes mediate complex, indirect interactions between species.

An impressive example of a positive, indirect effect of fleas on their host was highlighted by Tompkins (2011): "*Bartonella* spp. (a flea-vectored bacterial infection) declined in wood mice *Apodemus sylvaticus* in Ireland as densities of the invading bank vole *Myodes glareolus* increased. The infection was detected in wood mice, but not in bank voles, suggesting that the less competent invader acted as a sink for the parasite (Telfer et al. 2005b)". This invasive host-sink effect with fleas having positive effects on the endemic host is similar to what Mlynarek (2015) found with invasive dragonflies attracting both ectoparasites (mites) and endoparasites (gregarines) from the ecosystem, away from endemic dragonflies.

Another instructive example of a positive, indirect effect of fleas on other species are the fleas *Opisodasys keeni*, *Aetheca wagneri*, *Catallagia decipiens*, *Megabothris abantis*, *Peromyscopsylla selenis*, *Amaradix euphorbi*, *Peromyscopsylla hesperomys*, *Epitedia wenmanni*, *Malaraeus telchinus*, and *Rhadinopsylla fraterna*, that may help re-establish a forest's trees after fires. By parasitizing the teeming masses of deer mice (*Peromyscus* spp.), fleas appear to regulate populations of deer mice and oppose the mice's destruction of an area's seed bank. This, indirectly, benefits plants and allows trees to regrow (Zwolak et al. 2013). These results suggest that endemic parasites may act to protect communities from invading or irrupting host species and stabilize endemic diversity.

The "fleas to trees" hypothesis is untested, although preliminary results are consistent with its implications. Zwolak et al. (2013) notes that effects of fleas that are not disease vectors on ecosystems are "largely unstudied".

One of the best-known experimental systems on fleas that are not vectors is of Great Tit (*Parus major*) nests, where in one study, Heeb et al. (2000) manipulated flea infestation levels (the flea was *Ceratophyllus gallinae*) and nest humidity, and claimed the first experimental results showing: 1) "an ectoparasite acting as a physical ecosystem engineer where nest humidity increased as a result of flea infestations", 2) "an abiotic factor within nests can affect ectoparasite infracommunity structure", and 3) "an interspecific association between two ectoparasite species" (the other ectoparasite was *Protocalliphora* flies). Similar interspecific, often positive, co-occurrences were found with deer-mice parasites (Veitch et al. 2020).

Fleas that are disease vectors are known to affect ecosystem processes strongly, shown strikingly by experiments with insecticides and plague vaccines (Machett et al. 2010). Flea-borne plague bacterium (*Yersinia pestis*) is an invasive 'transformer species' of natural ecosystems (Eads and Biggins 2015). There is ambiguity over whether an invasive species can also be a keystone species, or only endemic species can count as keystone species. For my purposes, I will rely on the original

definition of "keystone species", considering species to be keystone if their presence or removal causes major effects in the community, regardless of whether the species is endemic or invasive.

Plague suppression treatments cause direct and indirect trophic effects on hosts, host predators, host food plants, and ancillary species. By causing periodic die-offs of prairie dogs (*Cynomys* spp.), fleas with plague can affect the entire ecosystem: nutrient cycling changes, such as carbon storage, nitrogen mineralization, and water infiltration; there are increases in graminoids, shrubs, and woody plants, but loss of nutritious forbs; burrows used by vertebrates and invertebrates for shelter are reduced; and prey is decreased for predators such as hawks (*Buteo*) and ferrets (*Mustela*). Prairie dogs remove foundation plants, fleas link prairie dogs with plague, plague-bacteria extirpate prairie dogs.

In conclusion, fleas can be keystone species. A flea can change its entire ecosystem.

Continued best wishes in regard to the coronavirus pandemic.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

Literature

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Zwolak, R., S. Meagher, J.W. Vaughn, S. Dziemian, and E.E. Crone. 2013. Reduced ectoparasite loads of deer mice in burned forest: From fleas to trees? *Ecosphere* 4: 1-10.

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Announcement

Terrestrial Parasite Tracker – Documentation of Arthropod Parasites of Vertebrates

<https://parasitetracker.org/>

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Featured Research

New Flea Species 2020

Hastriter, M. W. (2020). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua Province (Indonesia). Part V. *Astivalius*, *Idiochaetis*, *Muesebeckella*, and *Obtusifrontia* (Stivaliidae: Stivaliinae), with Description of Six New Species. *Annals of Carnegie Museum*, 86(2), 169-196.

(abstract in Flea News 86)

[A. archboldi, n. sp., A. mirzai, n. sp., A. toxopeusi, n. sp., I. rogersi, n. sp., Mu. niobiensis, n. sp., and

O. comohamulus, n. sp. Hastriter 2020]

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Keskin, A. (2020). A new flea species of the genus *Palaeopsylla* (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. *Journal of Medical Entomology*, 57(1), 88-91.

(abstract in Flea News 86)

[*Palaeopsylla* (*Palaeopsylla*) *beaucournui* Keskin 2020]

**

López-Berrizbeitia, M. F., Pérez, M. J., & Barquez, R. M. (2020). Description of two new species of *Ectinorus* (*Ectinorus*) (Siphonaptera, Rhopalopsyllidae) from Argentina, including a morphometric approach. *Acta Tropica*, 211, 105612.

(abstract in Flea News 86)

[*Ectinorus* (*Ectinorus*) *monicadiazae* n. sp. López-Berrizbeitia, Pérez and Barquez

Ectinorus (*Ectinorus*) *julianae* n. sp. Lopez-Berrizbeitia, Perez and Barquez]

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Nguyen Van Chau, and Nguyen Thu Van. 2004. Three new flea species (Insecta: Siphonaptera) collected on small mammals from Vietnam. *Vietnamese Academy of Science and Technology Journal of Biology* 26(3A): 19-26.

[*Medwayella vietnamensis* Chau and Van, 2004 on host *Menetes berdmorei* (Berdmore's ground squirrel) (Sciuridae)

Paraceras pagumani Chau and Van, 2004 on host *Paguma larvata* (masked palm civet) (Viverridae)

Paraceras berdmorei Chau and Van, 2004 on host *Menetes berdmorei*]

[Editor's note: Listed in Vietnam National Biodiversity Database as Pygiopsyllidae, but *Medwayella* is Stivaliidae in our Lewis database. *Paraceras* is Ceratophyllidae.]

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Zhang, Y., Shih, C., Rasnitsyn, A.P., Ren, D., Gao, T. (2020) A new flea from the Early Cretaceous of China. *Acta Palaeontologica Polonica*, 65, pp. 99-107. DOI: 10.4202/APP.00680.2019

Fleas are highly specialized holometabolic insects. So far, only 16 species of fossil fleas in five families have been reported due to the rare fossil records. At present, the earliest flea fossils are reported from the Middle Jurassic Jiulongshan Formation of Northeastern China. The descriptions of these earliest species pushed back the origin of Siphonaptera by at least 40 million years. It is generally accepted that saurophthirids are the "transitional" taxa from stem Jurassic fleas to living crown groups. Herein, we described a new "transitional" flea species, *Saurophthirus laevigatus* Zhang, Shih, Rasnitsyn, and Gao sp. nov., from the Lower Cretaceous Yixian Formation of Northeastern China, assigned to Saurophthiridae. The new species provides new evidence to support saurophthirids as a "transitional" group. Sexual dimorphism suggests significant differences in biology of opposite sexes in *Saurophthirus*. Analysis of described Mesozoic species demonstrates the body size reduction from the Middle Jurassic to the Early Cretaceous. Smaller body size was likely advantageous in reducing the probability of being detected and removed by the host and in minimizing flea's demand for blood intake and energy input, indicating the adaptation of the ectoparasitic lifestyle of fleas in their early stage of evolution.

[*Saurophthirus laevigatus* Zhang, Shih, Rasnitsyn, and Gao sp. nov.]

Taxonomy

Beaucournu J.-C., *Ordre des Siphonaptères (Puces)*, Chapter 37 in Aberlenc H., *Les Insectes du Monde* (2 volumes), Quae et Opie Ed., paru April 2021.

De Avelar, D.M.; Linardi, P.M. Siphonaptera. In: Andrade, H.T.A. & Serpa Filho, A. (org.). *Princípios Básicos de Entomologia Médica*. Natal, Brasil: Editora Caule de Papiro, 2021, p. 457-499.

Hastriter, M. W., & Eckerlin, R. P. (2020). Report of a Small Collection of *Kohlsia mojica* Tipton and Méndez, 1966 (Siphonaptera: Ceratophyllidae) from Costa Rica with a Discussion of Several Enigmatic Species of *Kohlsia* Traub, 1950. *Proceedings of the Entomological Society of Washington*, 122(3), 632-649.

(abstract in Flea News 86)

Kwak, M. L., & Hastriter, M. W. (2020). The Australian giant fleas *Macropsylla* Rothschild, 1905

(Siphonaptera: Macropsyllidae: Macropsyllinae), their identification, evolution, ecology, and conservation biology. *Systematic Parasitology*, 97(1), 107-118.

(abstract in *Flea News* 86)

LÓPEZ-BERRIZBEITIA, M. F., Sanchez, J., BARQUEZ, R., & Diaz, M. (2020). Taxonomic revision of the flea genus *Agastopsylla* Jordan & Rothschild 1923 (Siphonaptera: Ctenophthalmidae). *Anais da Academia Brasileira de Ciências*, 92(1).

Fleas of Argentina are receiving renewed systematic interest, but the identification of many species associated with small mammals can be problematic. We review the taxonomy of the flea genus *Agastopsylla* including the re-description of two species and one subspecies, and designate neotype and neallotype for *Agastopsylla hirsutior*, neotype for *Agastopsylla nylota nylota* from the “Colección Mamíferos Lillo Anexos” (CMLA), Universidad Nacional de Tucumán, Argentina, and neotype and neallotype for *Agastopsylla pearsoni* from the Natural History Museum (London, U.K.). Additionally, a key to identification of the species of *Agastopsylla* and a distribution map of the species of the genus are included.

Pawełczyk, O., Postawa, T., Blaski, M., Solarz, K. (2020) Morphology Reveals the Unexpected Cryptic Diversity in *Ceratophyllus gallinae* (Schrank, 1803) Infested *Cyanistes caeruleus* Linnaeus, 1758 Nest Boxes. *Acta Parasitologica*, 65 (4), pp. 874-881. DOI: 10.1007/s11686-020-00239-6

Purpose: The main aim of our study was to examine morphological differentiation between and within sex of hen fleas—*Ceratophyllus gallinae* (Schrank, 1803) population collected from Eurasian blue tit (*Cyanistes caeruleus* Linnaeus, 1758), inhabiting nest boxes and to determine the morphological parameters differentiating this population.

Methods: A total of 296 fleas were collected (148 females and 148 males), determined to species and sex, then the following characters were measured in each of the examined fleas: body length, body width, length of head, width of head, length of comb, height of comb, length of tarsus, length of thorax and length of abdomen.

Results: The comparison of body size showed the presence of two groups among female and male life forms of the hen flea, which mostly differed in length of abdomen, whereas the length of head and tarsus III were less variable.

Conclusion: Till now, the only certain information is the presence of two adult life forms of *C. gallinae*. The genesis of their creation is still unknown and we are not able to identify the mechanism responsible for the morphological differentiation of fleas collected from the same host. In order to find answer to this question, future research in the field of molecular taxonomy is required.

Zurita, A., García-Sánchez, Á. M., & Cutillas, C. (2020). *Ctenophthalmus baeticus boisseauorum* (Beaucournu, 1968) and *Ctenophthalmus apertus allani* (Smit, 1955)(Siphonaptera: Ctenophthalmidae) as synonymous taxa: morphometric, phylogenetic, and molecular characterization. *Bulletin of Entomological Research*, 1-14.

(abstract in Flea News 86)

Hosts and Distribution

Canto-Osorio, J.M., Cuxim-Koyoc, A., Ruiz-Piña, H.A., Morales-Malacara, J.B., Reyes-Novelo, E. (2020) Ectoparasites of *Didelphis virginiana* from Yucatan, Mexico. *Journal of Medical Entomology*, 57 (6), pp. 1821-1829. DOI: 10.1093/jme/tjaa106

A faunal study is presented to determine the species of ectoparasites found in the opossum *Didelphis virginiana* (Kerr) (Marsupialia: Didelphidae). For this, Tomahawk traps were placed in the peridomiciles of a rural town in Yucatán, to capture individuals of this marsupial and proceed to the collection of their ectoparasites. A total of 3,023 arthropods were collected from 145 opossums. The most frequent ectoparasites were the acarines *Ornithodoros (Alectorobius) nr. talaje* (64.8%) (Argasidae); *Ornithonyssus wernecki* Fonseca (53.8%) (Macronyssidae) and *Didelphiliculus serrifer* Fain (25.5%) (Atopomelidae); the ticks *Amblyomma parvum* Aragão (8.3%) and *A. mixtum* Koch (10.3%) (Ixodidae); and the fleas *Ctenocephalides felis* (Bouché) (20.7%) and *Pulex simulans* Baker (8.3%) (Pulicidae). It is concluded that the ectoparasite fauna of this marsupial is mainly composed of euryxenous organisms, which have been documented as vectors of diseases caused by rickettsial bacteria. Since *D. virginiana* is widely adapted to the peridomiciliary environment in the region, this study allows recognizing those that represent a potential risk for the transmission of vector-borne zoonotic diseases.

Cardoso, P., Barton, P. S., Birkhofer, K., Chichorro, F., Deacon, C., Fartmann, T., . . . Samways, M. J. (2020). Scientists' warning to humanity on insect extinctions. *Biological Conservation*, 242 doi:10.1016/j.biocon.2020.108426

Highlights

- We are pushing many ecosystems beyond recovery, resulting in insect extinctions.
- Causes are habitat loss, pollution, invasives, climate change, and overexploitation.
- We lose biomass, diversity, unique histories, functions, and interaction networks.
- Insect declines lead to loss of essential, irreplaceable services to humanity.

- Action to save insect species is urgent, for both ecosystems and human survival.

Abstract: Here we build on the manifesto ‘World Scientists’ Warning to Humanity, issued by the Alliance of World Scientists. As a group of conservation biologists deeply concerned about the decline of insect populations, we here review what we know about the drivers of insect extinctions, their consequences, and how extinctions can negatively impact humanity.

We are causing insect extinctions by driving habitat loss, degradation, and fragmentation, use of polluting and harmful substances, the spread of invasive species, global climate change, direct overexploitation, and co-extinction of species dependent on other species.

With insect extinctions, we lose much more than species. We lose abundance and biomass of insects, diversity across space and time with consequent homogenization, large parts of the tree of life, unique ecological functions and traits, and fundamental parts of extensive networks of biotic interactions. Such losses lead to the decline of key ecosystem services on which humanity depends. From pollination and decomposition, to being resources for new medicines, habitat quality indication and many others, insects provide essential and irreplaceable services. We appeal for urgent action to close key knowledge gaps and curb insect extinctions. An investment in research programs that generate local, regional and global strategies that counter this trend is essential. Solutions are available and implementable, but urgent action is needed now to match our intentions.

"A historical example of indirect effects of invasive species is the co-extinction of Christmas Island flea (*Xenopsylla nesiotus* Jordan & Rothschild, 1909), resulting from loss of the Christmas Island rat (*Rattus macleari* Thomas, 1887) due to the introduced black rat (*Rattus rattus* L., 1758) (Kwak, 2018)."

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Guzmán-Cornejo, C., Herrera-Mares, A., Ugalde-Medina, A., López-Pérez, A.M., Del Castillo-Martínez, L., Acosta-Gutiérrez, R., Cabrera-Garrido, M., Morales-Malacara, J.B. (2020) Arthropods Associated with Mammals. Their Importance as Part of the Richness in a Biosphere Reserve in Mexico. *Journal of Medical Entomology*, 57 (3), pp. 780-787. DOI: 10.1093/jme/tjz237

The Chamela Biological Station (ChBS) is located in the Pacific Coast of Mexico in the State of Jalisco. This represents one of the core areas of the Chamela-Cuixmala Biosphere Reserve, characterized by one of the most threatened ecosystems, the tropical dry forest. Although accumulated knowledge of biological diversity, only few studies have been focused on ectoparasites or ectodytes associated with mammals, only 23 arthropod taxa had been recorded. In order to increase knowledge about arthropods associated with Mexican mammals, the objective of this work was to record the richness of arthropods (mites, ticks, lice, and fleas) associated with small and medium-sized mammals in the ChBS. A total of 81 hosts belonging to four orders, six families and nine species were captured. From these hosts, 4,946 arthropods were recovered: 4,007 mites, 673 ticks, 230 lice, and 36 fleas. Among medium-sized mammals, *Nasua narica* (L.) and *Didelphis virginiana* Kerr showed the highest levels of richness, with six arthropod taxa; among rodents, *Heteromys pictus* (Thomas) had the highest

number of associated species (five). Within the 22 arthropod taxa registered in the present work, 12 represent new records for the reserve, and 3 represent new records for Mexico. With this study, the arthropod fauna associated with mammals in the ChBS has been raised to 38 taxa. In terms of biological conservation, knowledge of the species that inhabit natural reserves must be a priority, since this represents the baseline for species protected, not only in Mexico but around the world.

Kwak, M.L., Heath, A.C.G., Cardoso, P. (2020) Methods for the assessment and conservation of threatened animal parasites. *Biological Conservation*, 248, art. no. 108696. DOI: 10.1016/j.biocon.2020.108696

Animal parasites are species rich and ecologically diverse, some with complex life cycles and life histories and consequently many are vulnerable to extinction. While the conservation biology community is becoming cognisant of the plight of threatened parasites, coupled at times with potential host co-extinction, feasible criteria have not been developed to specifically assess their threat risk and methods to conserve them have also received limited attention. Methodologies and criteria are presented here to aid the assessment of the conservation status of threatened parasites in a way consistent with the criteria of the International Union for the Conservation of Nature (IUCN). Such assessments assist in prioritizing conservation efforts, a crucial task given parasites are integral to the functioning and survival of global ecosystems.

Light, J.E., Durden, L.A., Connor, B.M.O., Preisser, W.C., Acosta, R., Eckerlin, R.P. (2020) Checklist of ectoparasites of cricetid and heteromyid rodents in México. *Therya*, 11 (1), pp. 79-136. DOI: 10.12933/therya-20-785

We present an updated checklist of ectoparasite species (including mites, ticks, lice, and fleas) infesting cricetid and heteromyid rodents (Rodentia: Cricetidae and Heteromyidae) in México. For each parasite species, we include information on parasite taxonomy, host associations, collection localities, citations from the literature, and museum collections, when available. We recorded more than 172 mite, 114 flea, 22 tick, and 22 louse species from a minimum of 110 host species (82 cricetid and 28 heteromyid species) distributed across 31 states in México. This work represents the most up-to-date and comprehensive checklist of ectoparasite species parasitizing cricetid and heteromyid rodents in México.

López-Berrizbeitia, M.F., Acosta-Gutiérrez, R., Díaz, M.M. (2020) Fleas of mammals and patterns of distributional congruence in northwestern Argentina: A preliminary biogeographic analysis. *Heliyon*, 6 (9), art. no. e04871. DOI: 10.1016/j.heliyon.2020.e04871

In few groups of parasites have the patterns of distribution been studied using quantitative methods, even though, the study of these organisms indirectly provides information on the biogeographic history of their hosts, and in turn, the history of the hosts allows elucidation of

speciation events of the parasites. Our objective was to quantitatively identify distributional congruence patterns of native fleas in northwestern Argentina. We analyzed 159 georeferenced distributional records of 47 species and six subspecies of fleas in northwestern Argentina using NDM/VNDM software. We found eight consensus areas, defined by 17 species and two subspecies, included in six patterns of distributional congruence (PDCs) with endemic and non-endemic fleas. The PDCs with the greatest values of endemism (E) were mainly associated with Monte and Yungas Forests areas. All patterns indicated strong tendency of the Yungas Forests as a possible endemism area. Our results indicate that distributional congruence centers are generally located in Yungas Forests areas and highlight the importance of these areas in conservation and historical biology. This new information will allow delimitation of areas in the region at a more detailed resolution in the future.

Areas of endemism, Host, Monte, NDM/VNDM, Ectoparasites, Richness, Siphonaptera, Yungas forests, Computer science, Environmental science, Veterinary medicine, Health sciences

Ramírez-Chaves, H. E., Tamayo-Zuluaga, A. F., Henao-Osorio, J. J., Cardona-Giraldo, A., Ossa-López, P. A., & Rivera-Páez, F. A. (2020). The chiggerflea *Hectopsylla pulex* (Siphonaptera: Tungidae): Infestation on *Molossus molossus* (Chiroptera: Molossidae) in the central Andes of Colombia. *Zoologia* 37, 1-5. doi:10.3897/zoologia.37.e53092

Vanstreels, R.E.T., Palma, R.L., Mironov, S.V. (2020) Arthropod parasites of Antarctic and Subantarctic birds and pinnipeds: A review of host-parasite associations. *International Journal for Parasitology: Parasites and Wildlife*, 12, pp. 275-290. DOI: 10.1016/j.ijppaw.2020.03.007

Due to its cold and dry climate and scarcity of ice-free land, Antarctica has one of the most extreme environments on our planet. To survive in the Antarctic region, parasitic arthropods must either remain closely associated with their hosts throughout the entire life cycle or develop physiological adaptations to survive in the terrestrial habitat while their hosts are away foraging at sea or overwintering at lower latitudes. Forty-eight species of birds and seven species of pinnipeds breed in the Antarctic region, with 158 species/subspecies of parasitic arthropods recorded thus far, comprising: sucking lice (Echinophthiriidae), chewing lice (Menoponidae, Philopteridae), fleas (Ceratophyllidae, Pygiopsyllidae, Rhopalopsyllidae), pentastomes (Reighardiidae), hard ticks (Ixodidae), nest-associated haematophagous mites (Laelapidae), nasal mites (Halarachnidae, Rhinonyssidae) and feather mites (Alloptidae, Avenzoariidae, Xolalgidae, Freyanidae). In this review, we provide an updated compilation of the available information on the host-parasite associations of arthropods infesting birds and pinnipeds in the Antarctic region, and discuss some over-arching ecological patterns and gaps of knowledge.

Zajac, Z., Kulisz, J., Woźniak, A. (2020) Flea communities on small rodents in eastern Poland. *Insects*, 11 (12), art. no. 894, pp. 1-12. DOI: 10.3390/insects11120894

Fleas are hematophagous insects infesting mainly small mammals and, less frequently, birds.

With their wide range of potential hosts, fleas play a significant role in the circulation of pathogens in nature. Depending on the species, they can be vectors for viruses, bacteria, rickettsiae, and protozoa and a host for some larval forms of tapeworm species. The aim of this study was to determine the species composition of fleas and their small rodent host preferences in eastern Poland. Animals were captured in traps in various types of ecological habitats (a site covered by grassland vegetation within city limits, an unused agricultural meadow, and a fallow land near a mixed forest). The following rodent species were caught: *Apodemus agrarius*, *Apodemus flavicollis*, *Microtus arvalis*, and *Myodes glareolus*. Additionally, *Ctenophthalmus agyrtes*, *Ctenophthalmus assimilis*, *Hystrichopsylla talpae*, and *Nosopsyllus fasciatus* flea species were identified. The peak of the flea activity was noted in summer months. *C. agyrtes* was found to be the most abundant flea species in eastern Poland, while the greatest numbers of fleas were collected from the rodent species *A. agrarius*.

Physiology

Warburton, E.M., Khokhlova, I.S., Palme, R., Surkova, E.N., Van Der Mescht, L., Krasnov, B.R (2020) Flea infestation, social contact, and stress in a gregarious rodent species: Minimizing the potential parasitic costs of group-living. *Parasitology*, 147 (1), pp. 78-86. DOI: 10.1017/S0031182019001185

Both parasitism and social contact are common sources of stress that many gregarious species encounter in nature. Upon encountering such stressors, individuals secrete glucocorticoids and although short-term elevation of glucocorticoids is adaptive, long-term increases are correlated with higher mortality and deleterious reproductive effects. Here, we used an experimental host-parasite system, social rodents *Acomys cahirinus* and their characteristic fleas *Parapulex chephrenis*, in a fully-crossed design to test the effects of social contact and parasitism on stress during pregnancy. By analysing faecal glucocorticoid metabolites, we found that social hierarchy did not have a significant effect on glucocorticoid concentration. Rather, solitary females had significantly higher glucocorticoid levels than females housed in pairs. We found a significant interaction between the stressors of parasitism and social contact with solitary, uninfested females having the highest faecal glucocorticoid metabolite levels suggesting that both social contact and infestation mitigate allostatic load in pregnant rodents. Therefore, the increased risk of infestation that accompanies group-living could be outweighed by positive aspects of social contact within *A. cahirinus* colonies in nature.

Evolution

Álvarez-Parra, S., Delclòs, X., Solórzano-Kraemer, M.M., Alcalá, L., Peñalver, E. (2020) Cretaceous amniote integuments recorded through a taphonomic process unique to resins. *Scientific Reports*, 10 (1), art. no. 19840. DOI: 10.1038/s41598-020-76830-8

Fossil records of vertebrate integuments are relatively common in both rocks, as compressions, and amber, as inclusions. The integument remains, mainly the Mesozoic ones, are of great interest due

to the panoply of palaeobiological information they can provide. We describe two Spanish Cretaceous amber pieces that are of taphonomic importance, one bearing avian dinosaur feather remains and the other, mammalian hair. The preserved feather remains originated from an avian dinosaur resting in contact with a stalactite-shaped resin emission for the time it took for the fresh resin to harden. The second piece shows three hair strands recorded on a surface of desiccation, with the characteristic scale pattern exceptionally well preserved and the strands aligned together, which can be considered the record of a tuft. These assemblages were recorded through a rare biostratinomic process we call “pull off vestiture” that is different from the typical resin entrapment and embedding of organisms and biological remains, and unique to resins. The peculiarity of this process is supported by actualistic observations using sticky traps in Madagascar. Lastly, we reinterpret some exceptional records from the literature in the light of that process, thus bringing new insight to the taphonomic and palaeoecological understanding of the circumstances of their origins.

Kundrát, M., Rich, T. H., Lindgren, J., Sjövall, P., Vickers-Rich, P., Chiappe, L. M., & Kear, B. P. (2020). A polar dinosaur feather assemblage from Australia. *Gondwana Research*, 80, 1-11. doi:10.1016/j.gr.2019.10.004

Driscoll, T. P., Verhoeve, V. I., Brockway, C., Shrewsbury, D. L., Plumer, M., Sevdalis, S. E., . . . Gillespie, J. J. (2020). Evolution of *Wolbachia* mutualism and reproductive parasitism: Insight from two novel strains that co-infect cat fleas. *PeerJ*, 8 doi:10.7717/peerj.10646

Driscoll, T. P., Verhoeve, V. I., Gillespie, J. J., Johnston, J. S., Guillotte, M. L., Rennoll-Bankert, K. E., . . . Azad, A. F. (2020). A chromosome-level assembly of the cat flea genome uncovers rampant gene duplication and genome size plasticity. *BMC Biology*, 18(1) doi:10.1186/s12915-020-00802-7

Khokhlova, I.S., Fielden, L.J., Warburton, E.M., Van Der Mescht, L., Krasnov, B.R. (2020) Feeding performance on a novel host: No adaptation over generations and differential patterns in two flea species. *Parasitology*, 147 (6), pp. 721-728. DOI: 10.1017/S0031182020000244

To model the colonization of a novel host by fleas, *Synosternus cleopatrae* and *Xenopsylla ramesis*, we established experimental lines maintained for 15 generations on a principal or a novel host (either co-occurring with a flea or not). We compared the blood meal size and the energy expended for digestion by fleas from the 15th generation of each line on these hosts between hosts within a line and between lines within a host asking (a) whether fleas adapt to a novel host (increased blood consumption/decreased energy expended for digestion); (b) if yes, whether this adaptation leads to the loss of ability to exploit an original host, and (c) whether the success of adaptation to a novel host depends on its ecological co-occurrence with a flea. The blood consumption and digestion energetics of fleas fed on the principal host differed from those on other hosts. The effect of the principal host on feeding performance differed between fleas, with *S. cleopatrae* consuming less blood and expending more energy for digestion on the principal than on any other host, whereas the opposite was true for *X. ramesis*. No changes in feeding performance on a novel host over generations were found. We propose several explanations for the lack of adaptation to a novel host over time. We explain the poor performance of *S. cleopatrae* on its principal host via its immune response mounting pattern. We argue that the principal host of a parasite is not necessarily the host on which the parasite demonstrates the best performance.

Ecology

Ajith, Y., Dimri, U., Madhesh, E., Gopalakrishnan, A., Verma, M.R., Samad, H.A., Reena, K.K., Chaudhary, A.K., Devi, G., Bosco, J. (2020) Influence of weather patterns and air quality on ecological population dynamics of ectoparasites in goats. *International Journal of Biometeorology*, 64 (10), pp. 1731-1742. DOI: 10.1007/s00484-020-01952-7

Ectoparasitism has a damaging impact on the economy of goat production in India, but the factors influencing its distribution and dynamics are less explored. The present study was designed to investigate the influence of environmental factors like weather and air quality parameters on the occurrence of different types of ectoparasites in goats of two agro-climatic regions of India, viz. the Upper Gangetic Plain (UGP) and the Western Himalayas (WH). The prevalence survey for ectoparasitism among goats was conducted during the four distinct climatic seasons (winter, summer, monsoon, autumn) in both regions. The season-wise data of weather parameters (maximum and minimum temperature, relative humidity in morning and evening, sunrise and sunset time, mean daily temperature and relative humidity, daily variation in temperature and relative humidity, and day length) and air quality parameters (air quality index (AQI), particulate matter 2.5 μm (PM_{2.5}), particulate matter 10 μm (PM₁₀)) of both regions were analyzed in relation with the ectoparasitic prevalence pattern of corresponding regions. The results depict a noticeable correlation between the studied parameters and seasonal variation in the occurrence of each type of ectoparasites. This outcome on the interaction of studied parameters and ectoparasitism is intriguing and it opens a huge scope for future studies on the biometeorological aspects of host-parasite ecological interplay and evolutionary biology. The better understanding of climatological aspects of ectoparasite occurrences helps goat farmers in formulating appropriate timely intervention strategies for the economic control of ectoparasites, which in turn tackles ectoparasiticidal drug resistance and reduces threat of vector-borne diseases.

de Groot, M.D., Dumolein, I., Hiller, T., Sándor, A.D., Szentiványi, T., Schilthuizen, M., Aime, M.C., Verbeken, A., Haelewaters, D. (2020) On the fly: Tritrophic associations of bats, bat flies, and fungi. *Journal of Fungi*, 6 (4), art. no. 361, pp. 1-13. DOI: 10.3390/jof6040361

Parasitism is one of the most diverse and abundant modes of life, and of great ecological and evolutionary importance. Notwithstanding, large groups of parasites remain relatively understudied. One particularly unique form of parasitism is hyperparasitism, where a parasite is parasitized itself. Bats (Chiroptera) may be parasitized by bat flies (Diptera: Hippoboscoidea), obligate blood-sucking parasites, which in turn may be parasitized by hyperparasitic fungi, Laboulbeniales (Ascomycota: Laboulbeniomycetes). In this study, we present the global tritrophic associations among species within these groups and analyze their host specificity patterns. Bats, bat flies, and Laboulbeniales fungi are shown to form complex networks, and sixteen new associations are revealed. Bat flies are highly host-specific compared to Laboulbeniales. We discuss possible future avenues of study with regard to the dispersal of the fungi, abiotic factors influencing the parasite prevalence, and ecomorphology of the bat fly parasites.

Espinaze, M.P.A., Hui, C., Waller, L., Matthee, S. (2020) Nest-type associated microclimatic conditions as potential drivers of ectoparasite infestations in African penguin nests. *Parasitology Research*, 119

(11), pp. 3603-3616. DOI: 10.1007/s00436-020-06895-x

Nest design and characteristics can influence the microclimatic conditions in the nest. Nest-dwelling ectoparasites are sensitive to temperature and moisture and as such the conditions in the nest can influence parasite infestations. The endangered African penguin (*Spheniscus demersus*) breeds in different nest types and as yet little is known with regard to the microclimate and parasite infestation within these nests. This study characterized the microclimatic conditions in natural open, natural covered (with vegetation) and artificial nests, and assessed the relationship between nest characteristics (type, age, distance from the coast, orientation and entrance opening) and in-nest ectoparasite infestations and the health of African penguins in Stony Point, South Africa. Penguins (50 adults and 192 chicks) and their nests (n = 308) were sampled in 2016 and 2017. Soil temperature was higher in artificial than in natural nests, and soil and nest material moisture was lower in artificial and natural covered nests than natural open. Ectoparasite infestations were higher under warmer and drier conditions, in artificial nests and nests near the coastline. Penguin (adult and chick) body mass and chick body condition were lower in warmer nests and total plasma protein (in adults and chicks) was lower in drier nests. Given the potential adverse effects of ectoparasites on host species, it is recommended that conservation agencies implement a monitoring programme to assess the ectoparasite infestation in artificial nests across multiple colonies. This information will facilitate a more holistic penguin conservation management plan that may prevent further detrimental effects on this endangered penguin species.

Goldberg, A. R., Conway, C. J., & Biggins, D. E. (2020). Flea sharing among sympatric rodent hosts: Implications for potential plague effects on a threatened sciurid. *Ecosphere* 11(2) doi:10.1002/ecs2.3033

Gozzi, A. C., Lareschi, M., Navone, G. T., & Guichón, M. L. (2020). The enemy release hypothesis and *Callosiurus erythraeus* in Argentina: Combining community and biogeographical parasitological studies. *Biological Invasions*, 22(12), 3519-3531. doi:10.1007/s10530-020-02339-w

Krasnov, B. R., Stanko, M., Lareschi, M., & Khokhlova, I. S. (2020). Species co-occurrences in ectoparasite infracommunities: Accounting for confounding factors associated with space, time, and host community composition. *Ecological Entomology*, 45(5), 1158-1171. doi:10.1111/een.12900

Pathology and Control

Bosio, C.F., Jarrett, C.O., Scott, D.P., Fintzi, J., Hinnebusch, B.J. (2020) Comparison of the transmission efficiency and plague progression dynamics associated with two mechanisms by which fleas transmit *Yersinia pestis*. *PLoS Pathogens*, 16 (12), art. no. E1009092. DOI: 10.1371/journal.ppat.1009092

Yersinia pestis can be transmitted by fleas during the first week after an infectious blood meal, termed early-phase or mass transmission, and again after *Y. pestis* forms a cohesive biofilm in the flea foregut that blocks normal blood feeding. We compared the transmission efficiency and the progression of infection after transmission by *Oropsylla montana* fleas at both stages. Fleas were allowed to feed on mice three days after an infectious blood meal to evaluate early-phase transmission, or after they had

developed complete proventricular blockage. Transmission was variable and rather inefficient by both modes, and the odds of early-phase transmission was positively associated with the number of infected fleas that fed. Disease progression in individual mice bitten by fleas infected with a bioluminescent strain of *Y. pestis* was tracked. An early prominent focus of infection at the intradermal flea bite site and dissemination to the draining lymph node(s) soon thereafter were common features, but unlike what has been observed in intradermal injection models, this did not invariably lead to further systemic spread and terminal disease. Several of these mice resolved the infection without progression to terminal sepsis and developed an immune response to *Y. pestis*, particularly those that received an intermediate number of early-phase flea bites. Furthermore, two distinct types of terminal disease were noted: the stereotypical rapid onset terminal disease within four days, or a prolonged onset preceded by an extended, fluctuating infection of the lymph nodes before eventual systemic dissemination. For both modes of transmission, bubonic plague rather than primary septicemic plague was the predominant disease outcome. The results will help to inform mathematical models of flea-borne plague dynamics used to predict the relative contribution of the two transmission modes to epizootic outbreaks that erupt periodically from the normal enzootic background state.

Cui, Y., Schmid, B.V., Cao, H., Dai, X., Du, Z., Ryan Easterday, W., Fang, H., Guo, C., Huang, S., Liu, W., Qi, Z., Song, Y., Tian, H., Wang, M., Wu, Y., Xu, B., Yang, C., Yang, J., Yang, X., Zhang, Q., Jakobsen, K.S., Zhang, Y., Stenseth, N.C., Yang, R. (2020) Evolutionary selection of biofilm-mediated extended phenotypes in *Yersinia pestis* in response to a fluctuating environment. *Nature Communications*, 11 (1), art. no. 281. DOI: 10.1038/s41467-019-14099-w

Yersinia pestis is transmitted from fleas to rodents when the bacterium develops an extensive biofilm in the foregut of a flea, starving it into a feeding frenzy, or, alternatively, during a brief period directly after feeding on a bacteremic host. These two transmission modes are in a trade-off regulated by the amount of biofilm produced by the bacterium. Here by investigating 446 global isolated *Y. pestis* genomes, including 78 newly sequenced isolates sampled over 40 years from a plague focus in China, we provide evidence for strong selection pressures on the RNA polymerase ω -subunit encoding gene *rpoZ*. We demonstrate that *rpoZ* variants have an increased rate of biofilm production in vitro, and that they evolve in the ecosystem during colder and drier periods. Our results support the notion that the bacterium is constantly adapting—through extended phenotype changes in the fleas—in response to climate-driven changes in the niche.

Du, H., Zhu, A.-X., Wang, Y. (2020) Spatial prediction of flea index of transmitting plague based on environmental similarity. *Annals of GIS*, pp. 227-236. DOI: 10.1080/19475683.2020.1788639

Plague is a natural infectious disease which both people and mammals can contract once bitten by infected parasitic fleas. Flea index is an important outbreak indication. Outbreaks are more likely to occur for areas where flea index values are high. Thus, Prediction of spatial variation in flea index is vital to the prevention and control of the related plague. Existing methods of spatial prediction techniques require extensive field data to develop the needed models and limited for areas with insufficient field data. This paper presents an approach based on environmental similarity. The basic idea is that the more similar the environment of a location to locations at which the flea index is high the more likely that the flea index at this location is high as well. A method based on this idea was developed to predict spatial distribution of flea index of plague based on environmental similarity. A case study of mapping the flea index in Mongolian gerbil was conducted to evaluate the effectiveness of this idea and the environmental similarity method. The result showed that compared with multiple

linear regression (MLR), the environmental similarity method produced a flea index map of higher accuracy (root mean squared error (RMSE) of 6.6493 and mean absolute error (MAE) 1.2713 than that from MLR (7.293 RMSE and MAE for MLR is not 1.2713 but 2.928). In addition, the spatial distribution of prediction uncertainty estimated from the environmental similarity method can be used as an indicator of spatial distribution of prediction accuracy.

Elzinga, D.C., Stowe, S.R., Leland Russell, F. (2020) Modeling control methods to manage the sylvatic plague in black-tailed prairie dog towns. *Natural Resource Modeling*, 33 (2), art. no. e12255. DOI: 10.1111/nrm.12255

Black-tailed prairie dogs (*Cynomys ludovicianus*) are a keystone species of shortgrass and midgrass prairies, and they are essential for successful reintroduction efforts of endangered black-footed ferrets (*Mustela nigripes*). Sylvatic plague extirpates black-tailed prairie dog colonies. We present a model to determine optimal control methods to sustain a black-tailed prairie dog colony against the plague, including a recently developed vaccine. Our host submodel is a susceptible, exposed, infectious, vaccinated model, and our vector submodel is a susceptible, exposed, early-stage and late-stage infectious model with questing and on-host vectors. Both submodels are hybrid ordinary differential and difference equation models to reflect the phenology of black-tailed prairie dogs. Our model suggests vaccination efforts can substantially outperform previous control methods (e.g. reactive insecticidal dusting). Our model suggests that a vaccine that provides 18 months of immunity, administered annually, starting one year before an outbreak, can efficiently sustain a black-tailed prairie dog colony against the plague.

Recommendations for resource managers

The current sylvatic plague vaccine, administered annually, is insufficient to protect a black-tailed prairie dog colony against the sylvatic plague. Developing a vaccine with an immunity period of 18 months is critical for the persistence of a colony against the plague. While insecticidal dusting appears unable to assist in helping a black-tailed prairie dog colony recover from a plague outbreak, insecticidal dusting is still a critical prevention tool as plague transmission from the prairie dog flea is capable of causing a plague outbreak by itself. Black-tailed prairie dog colony recovery efforts appear sensitive to regularly administered culling. We recommend that colonies that are designated as part of the black-footed ferret recovery program be protected from population losses from recreational hunting and poisoning.

Erkunt Alak, S., Köseoğlu, A. E., Kandemir, Ç., Taşkın, T., Demir, S., Döşkaya, M., . . . Can, H. (2020). High frequency of knockdown resistance mutations in the para gene of cat flea (*Ctenocephalides felis*) samples collected from goats. *Parasitology Research*, 119(7), 2067-2073. doi:10.1007/s00436-020-06714-3

Liccioli, S., Stephens, T., Wilson, S.C., McPherson, J.M., Keating, L.M., Antonation, K.S., Bollinger, T.K., Corbett, C.R., Gummer, D.L., Lindsay, L.R., Galloway, T.D., Shury, T.K., Moehrenschrager, A. (2020) Enzootic maintenance of sylvatic plague in Canada's threatened black-tailed prairie dog ecosystem. *Ecosphere*, 11 (5), art. no. e03138. DOI: 10.1002/ecs2.3138

Data paucity can seem to hinder science-based approaches to the conservation of imperiled species. Yet, even individually limited datasets can improve understanding and management of complex ecological systems when carefully integrated. We demonstrate this approach to gain first

insights on the transmission ecology of *Yersinia pestis* in Grasslands National Park (GNP), Canada, where both the bacterium and its rodent host, the nationally threatened black-tailed prairie dog (BTPD, *Cynomys ludovicianus*), reach the northern limit of their distribution in North America. Primarily flea-borne, *Y. pestis* causes sylvatic plague, a disease of exceptional relevance to both human health and wildlife conservation. We integrated data collected independently by multiple organizations in 2010–2017 across 17 BTPD colonies, where the species co-occur with Richardson's ground squirrels (RGS, *Uroditellus richardsonii*). Available data included estimates of BTPD density and occupancy from visual counts and colony mapping; information on flea distribution, abundance, and prevalence of infection with *Y. pestis* from burrow swabbing, animal combing, and PCR assays; and the response of these variables to deltamethrin application on BTPD colony sections. Our analyses suggest that sylvatic plague in GNP is maintained at an enzootic level (i.e., chronic presence affecting a low proportion of individuals) with no evidence of widespread mortality, at least partially due to reduced flea activity after spring (percentage of prevalence in burrows: April–May = 11.69–33.89%; June–September: 1.75–3.19%), low prevalence of *Y. pestis* in flea samples (95% CI = 0.42–2.27%), and relatively low BTPD densities. Nonetheless, reducing flea prevalence through insecticide application had a positive effect on BTPD abundance, suggesting that enzootic plague is causing chronic mortality. Because flea prevalence on hosts was higher following drier years and higher on RGS than on BTPD (26.69% vs. 3.27%), insecticide application may be particularly important during dry periods and may need to take RGS and their movements into consideration. Differences between flea communities sampled by burrow swabbing and host combing suggest that plague surveillance should integrate both methods. Effects of projected climate change on vector life cycles, flea community composition, and host–parasite interactions warrant continued monitoring and an adaptive approach to species recovery actions and plague mitigation measures

El Hamzaoui, Basma, Antonio Zurita, Cristina Cutillas, Philippe Parola. 2020. Fleas and flea-borne diseases of North Africa. *Acta Tropica* 211 (2020) 105627.
<https://doi.org/10.1016/j.actatropica.2020.105627>

(abstract in Flea News 86)

Swei, Andrea, Lisa I. Couper, Lark L. Coffey, Durrell Kapan, and Shannon Bennett. 2020. Patterns, Drivers, and Challenges of Vector-Borne Disease Emergence. *VECTOR-BORNE AND ZOONOTIC DISEASES* Volume 20, Number 3, 159-170. DOI: 10.1089/vbz.2018.2432

Vector-borne diseases are emerging at an increasing rate and comprise a disproportionate share of all emerging infectious diseases. Yet, the key ecological and evolutionary dimensions of vector-borne disease that facilitate their emergence have not been thoroughly explored. This study reviews and synthesizes the existing literature to explore global patterns of emerging vector-borne zoonotic diseases (VBZDs) under changing global conditions. We find that the vast majority of emerging VBZDs are transmitted by ticks (Ixodidae) and mosquitoes (Culicidae) and the pathogens transmitted are dominated by Rickettsiaceae bacteria and RNA viruses (Flaviviridae, Bunyaviridae, and Togaviridae). The most common potential driver of these emerging zoonoses is land use change, but for many diseases, the driver is unknown, revealing a critical research gap. While most reported VBZDs are emerging in the northern latitudes, after correcting for sampling bias, Africa is clearly a region with the greatest share of emerging VBZD. We highlight critical gaps in our understanding of VBZD emergence and emphasize the importance of interdisciplinary research and consideration of deeper evolutionary processes to improve our capacity for anticipating where and how such diseases have and

will continue to emerge. Keywords: vector-borne, zoonotic, infectious disease, global change, tick, mosquito

[Editor's note: Fleas (primarily Hystrichopsyllidae) are shown in this paper as the 3rd most common vector for vector-borne zoonotic diseases, after ticks and mosquitoes. Surprisingly, Pulicidae is not the primary family of flea vectors according to this paper.]

Microbial Symbiotes

Brinkerhoff, R.J., Rinsland, H.S., Sato, S., Maruyama, S., Ray, C. (2020) Vector-Borne Pathogens in Ectoparasites Collected from High-Elevation Pika Populations. *EcoHealth*, 17 (3), pp. 333-344. DOI: 10.1007/s10393-020-01495-8

The American pika, *Ochotona princeps*, is projected to decline throughout North America as climate change reduces its range, and pikas have already disappeared from several locations. In addition to climate, disease spillover from lower elevation mammalian species might affect pikas. We sampled pika fleas in Colorado and Montana across elevations ranging from 2896 to 3612 m and screened them for the presence of DNA from rodent-associated bacterial pathogens (*Bartonella* species and *Yersinia pestis*) to test the hypothesis that flea exchange between pikas and rodents may lead to occurrence of rodent-associated pathogens in pika ectoparasites. We collected 275 fleas from 74 individual pikas at 5 sites in Colorado and one site in Montana. We found that 5.5% of 275 pika fleas in this study tested positive for rodent-associated *Bartonella* DNA but that variation in *Bartonella* infection prevalence in fleas among sites was not driven by elevation. Specifically, we detected DNA sequences from two loci (*gltA* and *rpoB*) that are most similar to *Bartonella grahamii* isolates collected from rodents in Canada. We did not detect *Y. pestis* DNA in our survey. Our results demonstrate evidence of rodent-associated flea-borne bacteria in pika fleas. These findings are also consistent with the hypothesis that rodent-associated pathogens could be acquired by pikas. Flea-borne pathogen spillover from rodents to pikas has the potential to exacerbate the more direct effects of climate that have been suggested to drive pika declines.

Melis, M., Espinoza-Carniglia, M., Savchenko, E., Nava, S., & Lareschi, M. (2020). Molecular detection and identification of *Rickettsia felis* in *Polygenis* (Siphonaptera, Rhopalopsyllidae, Rhopalopsyllinae) associated with cricetid rodents in a rural area from central Argentina. *Veterinary Parasitology: Regional Studies and Reports*, 21 doi:10.1016/j.vprsr.2020.100445

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**

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Fleas in Art and History

The Pied Piper of Hamelin

by Robert Browning

(published 1842 in *Bells and Pomengranates*)

(b. May 7, 1812, Camberwell, London, England - d. December 12, 1889, Venice, Kingdom of Sicily)

[Editor's note: the connection of fleas to Browning's poem is obscured by time, but the poem is inspired from an actual tale from 1284 in Hamelin, where flea-borne bubonic plague or *Rickettsia typhi* causing murine typhus may have been involved.

Dirckx, J.H. 1980. The pied Piper of Hamelin. A medical-historical interpretation. *Am. J. Dermatopathol.* 2(1): 39-45

Adam, D. 2000. Pied piper a plague risk? *Nature*, DOI 10.1038/news001019-11

Keeling, M.J. & C.A. Gilligan. 2000. Metapopulation dynamics of bubonic plague. *Nature* 407:903-906.)]

I

Hamelin Town's in Brunswick,
 By famous Hanover city;
 The river Weser, deep and wide,
 Washes its wall on the southern side;
 A pleasanter spot you never spied;
 But, when begins my ditty,
 Almost five hundred years ago,
 To see the townsfolk suffer so
 From vermin, was a pity.

II

Rats! They fought the dogs and killed the cats,
 And bit the babies in the cradles,
 And ate the cheeses out of the vats,
 And licked the soup from the cooks' own ladles,
 Split open the kegs of salted sprats,
 Made nests inside men's Sunday hats,
 And even spoiled the women's chats
 By drowning their speaking
 With shrieking and squeaking
 In fifty different sharps and flats.

III

At last the people in a body
 To the town hall came flocking:
 "'Tis clear," cried they, 'our Mayor's a noddy;
 And as for our Corporation--shocking
 To think we buy gowns lined with ermine
 For dolts that can't or won't determine
 What's best to rid us of our vermin!
 You hope, because you're old and obese,
 To find in the furry civic robe ease?
 Rouse up, sirs! Give your brains a racking
 To find the remedy we're lacking,
 Or, sure as fate, we'll send you packing!"
 At this the Mayor and Corporation
 Quaked with a mighty consternation.

IV

An hour they sat in council,
 At length the Mayor broke silence:
 "For a guilder I'd my ermine gown sell,
 I wish I were a mile hence!
 It's easy to bid one rack one's brain--
 I'm sure my poor head aches again,
 I've scratched it so, and all in vain
 Oh for a trap, a trap, a trap!"
 Just as he said this, what should hap
 At the chamber door but a gentle tap?
 "Bless us," cried the Mayor, "what's that?"
 (With the Corporation as he sat,
 Looking little though wondrous fat;
 Nor brighter was his eye, nor moister
 Than a too-long-opened oyster,
 Save when at noon his paunch grew mutinous

For a plate of turtle, green and glutinous)
 "Only a scraping of shoes on the mat?
 Anything like the sound of a rat
 Makes my heart go pit-a-pat!"

V

"Come in!"--the Mayor cried, looking bigger:
 And in did come the strangest figure!
 His queer long coat from heel to head
 Was half of yellow and half of red
 And he himself was tall and thin,
 With sharp blue eyes, each like a pin,
 And light loose hair, yet swarthy skin,
 No tuft on cheek nor beard on chin,
 But lips where smiles went out and in--
 There was no guessing his kith and kin!
 And nobody could enough admire
 The tall man and his quaint attire.
 Quoth one: "It's as if my great-grandsire,
 Starting up at the Trump of Doom's tone,
 Had walked this way from his painted tombstone!"

VI

He advanced to the council-table:
 And, "Please your honors," said he, "I'm able,
 By means of a secret charm, to draw
 All creatures living beneath the sun,
 That creep or swim or fly or run,
 After me so as you never saw!
 And I chiefly use my charm
 On creatures that do people harm,
 The mole and toad and newt and viper;
 And people call me the Pied Piper."
 (And here they noticed round his neck
 A scarf of red and yellow stripe,
 To match with his coat of the self-same check;
 And at the scarf's end hung a pipe;
 And his fingers, they noticed, were ever straying
 As if impatient to be playing
 Upon this pipe, as low it dangled
 Over his vesture so old-fangled.)
 "Yet," said he, "poor piper as I am,
 In Tartary I freed the Cham,
 Last June, from his huge swarm of gnats;
 I eased in Asia the Nizam
 Of a monstrous brood of vampyre-bats:
 And as for what your brain bewilders--

If I can rid your town of rats
 Will you give me a thousand guilders?"
 "One? Fifty thousand!" was the exclamation
 Of the astonished Mayor and Corporation.

VII

Into the street the Piper stept,
 Smiling first a little smile,
 As if he knew what magic slept
 In his quiet pipe the while;
 Then, like a musical adept,
 To blow the pipe his lips he wrinkled,
 And green and blue his sharp eyes twinkled,
 Like a candle-flame where salt is sprinkled;
 And ere three shrill notes the pipe uttered,
 You heard as if an army muttered;
 And the muttering grew to a grumbling;
 And the grumbling grew to a mighty rumbling;
 And out of the houses the rats came tumbling.
 Great rats, small rats, lean rats, brawny rats,
 Brown rats, black rats, gray rats, tawny rats,
 Grave old plodders, gay young friskers,
 Fathers, mothers, uncles, cousins,
 Cocking tails and pricking whiskers,
 Families by tens and dozens,
 Brothers, sisters, husbands, wives--
 Followed the Piper for their lives.
 From street to street he piped advancing,
 And step for step they followed dancing,
 Until they came to the river Weser
 Wherein all plunged and perished!
 ‹Save one who, stout as Julius Caesar,
 Swam across and lived to carry
 (As the manuscript he cherished)
 To Rat-land home his commentary:
 Which was, "At the first shrill notes of the pipe,
 I heard a sound as of scraping tripe,
 And putting apples, wondrous ripe,
 Into a cider-press's gripe:
 And a moving away of pickle-tub-boards,
 And a leaving ajar of conserve-cupboards,
 And a drawing the corks of train-oil-flasks,
 And a breaking the hoops of butter-casks:
 And it seemed as if a voice
 (Sweeter far than by harp or by psaltery
 Is breathed) called out, 'Oh rats, rejoice!
 The world is grown to one vast dry-saltery!
 So munch on, crunch on, take your nuncheon,

Breakfast, supper, dinner, luncheon!
 And just as a bulky sugar-puncheon,
 All ready staved, like a great sun shone
 Glorious scarce an inch before me,
 Just as methought it said 'Come bore me!
 -- I found the Weser rolling o'er me."

VIII

You should have heard the Hamelin people
 Ringing the bells till they rocked the steeple.
 Go," cried the Mayor, "and get long poles!
 Poke out the nests and block up the holes!
 Consult with carpenters and builders
 And leave in our town not even a trace
 Of the rats!"-- when suddenly, up the face
 Of the Piper perked in the market-place,
 With a, "First, if you please, my thousand guilders!"

IX

A thousand guilders! The Mayor looked blue;
 So did the Corporation too.
 For council dinners made rare havoc
 With Claret, Moselle, Vin-de-Grave, Hock;
 And half the money would replenish
 Their cellar's biggest butt with Rhenish.
 To pay this sum to a wandering fellow
 With a gypsy coat of red and yellow!
 "Beside," quoth the Mayor with a knowing wink,
 "Our business was done at the river's brink;
 We saw with our eyes the vermin sink,
 And what's dead can't come to life, I think.
 So, friend, we're not the folks to shrink
 From the duty of giving you something for drink,
 And a matter of money to put in your poke;
 But as for the guilders, what we spoke
 Of them, as you very well know, was in joke.
 Beside, our losses have made us thrifty
 . A thousand guilders! Come, take fifty!

X

The Piper's face fell, and he cried,
 "No trifling! I can't wait! Beside,
 I've promised to visit by dinnertime
 Bagdad, and accept the prime
 Of the Head-Cook's pottage, all he's rich in,
 For having left, in the Caliph's kitchen,

Of a nest of scorpions no survivor--
 With him I proved no bargain-driver,
 With you, don't think I'll bate a stiver!
 And folks who put me in a passion
 May find me pipe to another fashion."

XI

"How?" cried the Mayor, "d'ye think I brook
 Being worse treated than a Cook?
 Insulted by a lazy ribald
 With idle pipe and vesture piebald?
 You threaten us, fellow? Do your worst,
 Blow your pipe there till you burst!"

XII

Once more he stept into the street
 And to his lips again
 Laid his long pipe of smooth straight cane;
 And ere he blew three notes (such sweet
 Soft notes as yet musician's cunning
 Never gave the enraptured air)
 There was a rustling that seemed like a bustling
 Of merry crowds justling at pitching and hustling,
 Small feet were pattering, wooden shoes clattering,
 Little hands clapping, and little tongues chattering,
 And, like fowls in a farm-yard when barley is scattering,
 Out came the children running.
 All the little boys and girls,
 With rosy cheeks and flaxen curls,
 And sparkling eyes and teeth like pearls,
 Tripping and skipping, ran merrily after
 The wonderful music with shouting and laughter.

XIII

The Mayor was dumb, and the Council stood
 As if they were changed into blocks of wood,
 Unable to move a step or cry,
 To the children merrily skipping by--
 And could only follow with the eye
 That joyous crowd at the Piper's back.
 But how the Mayor was on the rack
 And the wretched Council's bosoms beat,
 As the Piper turned from the High Street
 To where the Weser rolled its water's
 Right in the way of their sons and daughters!

However he turned from South to West
 And to Koppelberg Hill his steps addressed,
 And after him the children pressed;
 Great was the joy in every breast.
 "He never can cross that mighty top!
 He's forced to let the piping drop
 And we shall see our children stop!
 When, lo, as they reached the mountain-side,
 A wondrous portal opened wide,
 As if a cavern was suddenly hollowed;
 And the Piper advanced and the children followed,
 And when all were in to the very last,
 The door in the mountain-side shut fast.
 Did I say all? No! One was lame,
 And could not dance the whole of the way;
 And in after years, if you would blame
 His sadness, he was used to say,--
 "It's dull in our town since my playmates left!
 I can't forget that I'm bereft
 Of all the pleasant sights they see,
 Which the Piper also promised me.
 For he led us, he said, to a joyous land,
 Joining the town and just at hand,
 Where waters gushed and fruit-trees grew,
 And flowers put forth a fairer hue,
 And everything was strange and new;
 The sparrows were brighter than peacocks here,
 And their dogs outran our fallow deer,
 And honey-bees had lost their stings,
 And horses were born with eagles' wings:
 And just as I became assured
 My lame foot would be speedily cured,
 The music stopped and I stood still,
 And found myself outside the hill,
 Left alone against my will,
 To go now limping as before,
 And never hear of that country more!

XIV

Alas, alas for Hamelin!
 There came into many a burgher's pate
 A text which says that heaven's gate
 Opens to the rich at as easy rate
 As the needle's eye takes a camel in!
 The mayor sent East, West, North and South,
 To offer the Piper, by word of mouth
 Wherever it was men's lot to find him,
 Silver and gold to his heart's content,

If he'd only return the way he went,
 And bring the children behind him.
 But when they saw 'twas a lost endeavor,
 And Piper and dancers were gone forever,
 They made a decree that lawyers never
 Should think their records dated duly
 If, after the day of the month and year,
 These words did not as well appear:
 "And so long after what happened here
 On the twenty-second of July,
 Thirteen hundred and seventy-six;"
 And the better in memory to fix
 The place of the children's last retreat,
 They called it the Pied Piper's Street,
 Where any one playing on pipe or tabor
 Was sure for the future to lose his labor.
 Nor suffered they hostelry or tavern
 To shock with mirth a street so solemn,
 But opposite the place of the cavern
 They wrote the story on a column,
 And on the great church-window painted
 The same, to make the world acquainted
 How their children were stolen away,
 And there it stands to this very day.
 And I must not omit to say
 That, in Transylvania there's a tribe
 Of alien people who ascribe
 To the outlandish ways and dress
 On which their neighbors lay such stress,
 To their fathers and mothers having risen
 Out of some subterranean prison
 Into which they were trepanned
 Long time ago in a mighty band
 Out of Hamelin town in Brunswick land,
 But how or why they don't understand.

XV

So, Willy, let you and me be wipers
 Of scores out with all men--especially pipers!
 And, whether they pipe us free, from rats or from mice,
 If we've promised them ought, let us keep our promise.

Anonymous. c. 1820. *Extraordinary Exhibition of the Industrious Fleas* (Handbills), New York
Public Library digital collections [nypl.digitalcollections.510d47e1-2a51-a3d9-e040-e00a18064a99.001.w (1)].






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Showing the condition of Flea-biting, and the means by which the Flea
begins, begins of youth, and afterwards age, are inserted into the human
pore, and gradual action that distinguishes the disease of this celebrated
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HANDBILLS OF AN EXHIBITION OF PERFORMING FLEAS, ABOUT 1820.

PCM V2C1893

Insects - Flea

Siphonaptera Literature (2020)

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